

## HARSTENA, Sweden

WMO: 025630

Lat: 58.251N

Lon: 17.011E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -10.8      | -8.4 | -15.9                         | 0.9 | -6.0 | -12.2 | 1.3 | -6.9 | 12.7                  | -0.2 | 11.4 | -0.6 | 4.2                   | 330  | 0.630 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.2                    | 24.6            | 18.9 | 23.2 | 18.2 | 21.9 | 17.5 | 20.3                | 23.1 | 19.4 | 22.0 | 18.4 | 21.0 | 3.6                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.2                            | 14.0 | 21.8 | 18.3 | 13.2 | 21.0 | 17.3 | 12.4 | 20.2 | 58.4          | 23.2 | 55.2 | 22.0 | 52.3 | 21.1 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.2          | 9.8  | 8.8 | -12.5                      | 27.5 | 3.7                | 1.5 | -15.1                                              | 28.5 | -17.3      | 29.4 | -19.4      | 30.3 | -22.0      | 31.4 |    |    |
| (5)               |      |     | -12.9                      | 21.3 | 3.6                | 1.2 | -15.5                                              | 22.1 | -17.7      | 22.9 | -19.7      | 23.6 | -22.3      | 24.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.0    | -0.6  | -0.6  | 1.9   | 5.4   | 10.2  | 15.0  | 18.4  | 17.9  | 13.9  | 8.5   | 4.4   | 1.4  |
|                                                                                  | DBStd       | 7.37   | 3.72  | 3.68  | 3.47  | 2.67  | 2.80  | 2.38  | 2.43  | 2.41  | 2.58  | 2.92  | 2.91  | 3.91 |
|                                                                                  | HDD10.0     | 1549   | 328   | 297   | 253   | 139   | 33    | 1     | 0     | 0     | 3     | 62    | 169   | 265  |
|                                                                                  | HDD18.3     | 3819   | 586   | 530   | 511   | 388   | 253   | 104   | 30    | 36    | 134   | 303   | 419   | 523  |
|                                                                                  | CDD10.0     | 832    | 0     | 0     | 1     | 2     | 38    | 149   | 259   | 246   | 119   | 17    | 1     | 0    |
|                                                                                  | CDD18.3     | 58     | 0     | 0     | 0     | 0     | 0     | 3     | 31    | 24    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 120    | 0     | 0     | 0     | 0     | 1     | 8     | 67    | 44    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 9      | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 3     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.5    | 5.5   | 5.0   | 4.5   | 4.0   | 3.8   | 3.9   | 3.6   | 3.8   | 4.4   | 4.8   | 5.3   | 5.3  |
| Precipitation                                                                    | PrecAvg     | 557    | 38    | 30    | 28    | 31    | 40    | 51    | 63    | 61    | 50    | 52    | 54    | 49   |
|                                                                                  | PrecMax     | 693    | 72    | 52    | 53    | 69    | 75    | 93    | 139   | 141   | 106   | 112   | 106   | 84   |
|                                                                                  | PrecMin     | 453    | 6     | 16    | 9     | 6     | 17    | 15    | 12    | 1     | 17    | 2     | 9     | 17   |
|                                                                                  | PrecStd     | 68     | 18    | 9     | 13    | 17    | 14    | 22    | 32    | 33    | 20    | 25    | 22    | 20   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.3   | 8.7   | 13.5  | 15.3  | 19.9  | 24.2  | 27.3  | 26.7  | 21.5  | 16.3  | 11.7  | 11.0 |
|                                                                                  |             | MCWB   | 6.3   | 5.5   | 8.0   | 10.0  | 13.4  | 15.9  | 19.9  | 19.7  | 16.2  | 13.0  | 9.6   | 8.8  |
|                                                                                  | 2%          | DB     | 6.2   | 6.6   | 10.1  | 12.7  | 17.5  | 21.9  | 25.1  | 24.5  | 19.9  | 14.5  | 10.3  | 8.6  |
|                                                                                  |             | MCWB   | 4.5   | 4.3   | 5.9   | 8.1   | 12.3  | 14.8  | 19.4  | 19.1  | 15.8  | 12.1  | 8.7   | 6.7  |
|                                                                                  | 5%          | DB     | 5.0   | 5.3   | 8.0   | 11.0  | 15.9  | 20.3  | 23.6  | 23.0  | 18.6  | 13.4  | 9.1   | 7.0  |
|                                                                                  |             | MCWB   | 3.7   | 3.4   | 4.8   | 7.2   | 11.5  | 14.2  | 18.7  | 18.4  | 15.3  | 11.5  | 7.6   | 5.6  |
|                                                                                  | 10%         | DB     | 3.9   | 3.9   | 6.3   | 9.4   | 14.4  | 18.9  | 22.4  | 21.6  | 17.4  | 12.4  | 8.0   | 5.9  |
|                                                                                  |             | MCWB   | 3.0   | 2.6   | 3.8   | 6.3   | 10.8  | 13.7  | 18.2  | 17.7  | 14.7  | 10.8  | 6.8   | 4.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.6   | 6.2   | 8.7   | 10.9  | 14.6  | 17.7  | 21.9  | 22.1  | 17.6  | 14.0  | 10.4  | 8.9  |
|                                                                                  |             | MCDB   | 7.9   | 8.1   | 12.9  | 14.7  | 17.7  | 22.2  | 24.6  | 24.7  | 19.9  | 15.5  | 11.1  | 10.8 |
|                                                                                  | 2%          | WB     | 4.9   | 4.6   | 6.5   | 8.9   | 13.2  | 16.5  | 20.7  | 20.3  | 16.7  | 12.7  | 8.9   | 6.9  |
|                                                                                  |             | MCDB   | 5.8   | 6.2   | 9.2   | 11.6  | 16.5  | 20.2  | 23.7  | 22.8  | 18.8  | 13.9  | 10.1  | 8.0  |
|                                                                                  | 5%          | WB     | 3.9   | 3.6   | 5.2   | 7.6   | 12.2  | 15.6  | 19.7  | 19.3  | 15.9  | 11.9  | 7.9   | 5.9  |
|                                                                                  |             | MCDB   | 4.7   | 4.9   | 7.5   | 10.3  | 15.2  | 18.9  | 22.6  | 21.7  | 17.8  | 13.1  | 8.8   | 6.8  |
|                                                                                  | 10%         | WB     | 3.0   | 2.8   | 4.1   | 6.6   | 11.2  | 14.8  | 18.9  | 18.4  | 15.1  | 11.0  | 7.1   | 5.0  |
|                                                                                  |             | MCDB   | 3.9   | 3.8   | 5.9   | 9.0   | 13.8  | 17.8  | 21.6  | 20.9  | 16.9  | 12.3  | 7.8   | 5.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.5   | 3.9   | 5.0   | 5.7   | 6.1   | 6.3   | 6.2   | 5.8   | 4.9   | 3.8   | 3.1   | 3.4  |
|                                                                                  |             | MCDBR  | 4.4   | 5.1   | 7.5   | 8.0   | 8.0   | 8.2   | 7.7   | 7.1   | 6.3   | 5.0   | 4.4   | 4.5  |
|                                                                                  |             | MCWBR  | 3.8   | 3.7   | 4.7   | 4.6   | 4.3   | 4.1   | 3.8   | 3.8   | 3.8   | 3.5   | 3.9   | 4.0  |
|                                                                                  | 5% WB       | MCDBR  | 4.0   | 4.7   | 7.0   | 7.6   | 7.5   | 7.3   | 6.6   | 6.1   | 5.8   | 4.5   | 3.9   | 4.1  |
|                                                                                  |             | MCWBR  | 3.7   | 3.6   | 4.7   | 4.8   | 4.6   | 4.2   | 3.5   | 3.8   | 3.8   | 3.5   | 3.7   | 3.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.292  | 0.325 | 0.341 | 0.379 | 0.375 | 0.364 | 0.394 | 0.384 | 0.353 | 0.342 | 0.318 | 0.279 |      |
|                                                                                  | taud        | 2.272  | 2.324 | 2.366 | 2.322 | 2.408 | 2.473 | 2.410 | 2.429 | 2.496 | 2.424 | 2.291 | 2.200 |      |
|                                                                                  | Ebn at Noon | 551    | 704   | 805   | 827   | 859   | 874   | 835   | 817   | 791   | 680   | 505   | 431   |      |
|                                                                                  | Edh at Noon | 46     | 70    | 88    | 108   | 105   | 100   | 105   | 96    | 78    | 63    | 46    | 36    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.34   | 0.91  | 2.42  | 4.04  | 5.38  | 5.91  | 5.40  | 4.31  | 2.89  | 1.29  | 0.45  | 0.24  |      |
|                                                                                  | RadStd      | 0.05   | 0.17  | 0.23  | 0.41  | 0.42  | 0.26  | 0.49  | 0.38  | 0.34  | 0.20  | 0.09  | 0.03  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (50 neighbors) | +0.39 | N/A     | N/A    | N/A     | N/A   | N/A   | -122        | -157    | N/A     | N/A     |

Nomenclature: See separate page